

Tahir Mehmood

+92-349-9577864 — tahirmehmoodrehmani@gmail.com — linkedin.com/in/tahir-mehmood — github.com/tahirmehmood23423

Professional Summary

AI/ML Engineer specializing in NLP, Transformer architectures, and RAG systems. Experienced in end-to-end ML pipeline development and deployment for healthcare and legal applications using PyTorch, FastAPI, and Docker. Adept at collaborating in research and academic environments, with a strong foundation in deep learning, federated systems, and real-world AI deployment.

Technical Skills

ML & AI: NLP, Transformers, RAG, CNN, LSTM, Attention Mechanisms, Federated Learning, Time-Series Modeling

Languages: Python, C++

Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, NLTK, LangChain

Deployment: FastAPI, Streamlit, Docker, REST APIs

Tools: Git, GitHub, SQL, Azure, Jupyter Notebook, VS Code

Experience

Coding Fellow Sep 2024 – Present

Murabbi, NUST — Islamabad

- Taught programming and AI concepts to undergraduates through structured sessions and hands-on projects.

Teaching Assistant — Machine Learning Sep 2025 – Dec 2025

NUST — Islamabad

- Delivered tutorials on supervised learning, model evaluation, and feature engineering; supported student debugging.

Research Assistant — Deep Learning Jan 2024 – Dec 2024

NUST — Islamabad

- Built and optimized DL pipelines for time-series data; handled preprocessing, training, tuning, and evaluation.

Projects

LLM-Based Legal Chatbot *PyTorch, FastAPI, Streamlit, Docker, LangChain* Sep 2024 – Present

- Built a RAG system with LangChain and vector DB for real-time legal query answering.
- Developed RESTful APIs with FastAPI and containerized the full stack using Docker.

Parkinson's Disease Classification *PyTorch, CNN-LSTM-Attention* Jan – Dec 2024

- Classified Parkinson's stages from IMU sensor data (469 subjects); achieved **91% accuracy** and **94% F1-score**.
- Optimized model for real-time inference on embedded systems.

Federated Learning for Leukemia Detection *Swin Transformer, PyTorch* Jan – Dec 2024

- Designed a privacy-preserving federated framework across distributed medical datasets using Swin Transformer.

Education

MS Computer Science 2024 – 2026

NUST, Islamabad

BE Electrical Engineering 2019 – 2023

NUST, Islamabad

Certifications

- Data Science Specialization — Coursera

Leadership & Volunteering

Director — Alif School of Excellence

- Oversaw academic operations, curriculum delivery, and teaching resources for a non-profit school.